

INITIAL VALUE PROBLEM

Some time It is required to find
The soln of D.E subjected to
Supplementary Conditions. If

The Condition relate to one Value
of the Independent Variable.

such as $y(x_0) = y_0$

and $y'(x_0) = y_1$

These are Initial Value Problem.

Ans
$$y = (A)x^2 + (B)x$$
$$x=1, y=2$$
$$2 = A + B$$

$$y' = 2Ax + B$$

$$1 = 2A + B$$
$$A = -1, B = 3$$

$$y = -x^2 + 3x$$

$$y(1) = 2$$
$$y'(1) = 1$$

$$y'(1) = 1$$

BOUNDARY VALUE PROBLEM:

The Problem of finding the soln of D.E such that all the associated supplementary condition relate to the two different values of the Independent Variable is called the two point boundary value problem. OR Simple Boundary Problem e.g

$$y(x_0) = y_0$$

$$y'(x_1) = y_1$$

$$2 = A + B$$

$$1 = 2A + B$$

$$\boxed{1 = -A} \quad A = -1$$

$$2 = -1 + B$$

$$B = 3$$

CHAPTER No. 9
EXERCISE No 9.1

Q2 Find D.E of which
The given function is soln.

(i) $y = x + 3e^{-x}$ ——— ①

Diff w.r.t x

$$\frac{dy}{dx} = 1 + 3e^{-x}(-1)$$

$$\frac{dy}{dx} = 1 - 3e^{-x} \text{ ——— ②}$$

adding ①, ②

$$\frac{dy}{dx} + y = 1 + x.$$

is The Required D.E

$$\frac{dy}{dx} = -3(y e^{3x}) e^{-3x} + 3x^2 e^{-3x}$$

$$= -3y \left(e^{3x-3x} \right) + 3x^2 e^{-3x}$$

$$\frac{dy}{dx} = -3y + 3x^2 e^{-3x}$$

$$e^0 = 1$$

Q2 Find D.E of which
The given function is soln.

$$(ii) \quad y = (x^3 + c) e^{-3x} \quad \text{--- (1)}$$

Diff w. r. t. x (By using Product Rule)

$$\frac{dy}{dx} = (x^3 + c) \cdot \frac{d}{dx} e^{-3x} + e^{-3x} \cdot \frac{d}{dx} (x^3 + c)$$

$$= (x^3 + c) e^{-3x} \cdot \frac{d}{dx} (-3x) + e^{-3x} [3x^{3-1} \cdot 1 + 0]$$

$$\frac{dy}{dx} = -3(x^3 + c) e^{-3x} + 3x^2 e^{-3x} \quad \text{--- (2)}$$

$$\text{From (1)} \quad (x^3 + c) = \frac{y}{e^{-3x}}$$

$$(x^3 + c) = y e^{3x} \quad \text{Put in (2)}$$

$$\frac{dy}{dx} = -3(y e^{3x}) e^{-3x} + 3x^2 e^{-3x}$$

$$= -3y \left(e^{3x-3x} \right) + 3x^2 e^{-3x}$$

$$\frac{dy}{dx} = -3y + 3x^2 e^{-3x} \quad \text{Ans}$$

$$e^0 = 1$$

CHAPTER No. 9
EXERCISE NO 9.1

Q2 Find D.E of which
The given function is soln.

(ii) $ax + \ln y = y + b$ — ①

Diff it w.r.t x

$$a(1) + \frac{1}{y} \cdot \frac{dy}{dx} = \frac{dy}{dx} + 0$$

$$a + \frac{1}{y} \cdot \frac{dy}{dx} = \frac{dy}{dx}$$

$$\textcircled{a} + \bar{y}^{-1} \cdot \frac{dy}{dx} = \frac{dy}{dx}$$

Diff w.r.t. x

$$0 + \bar{y}^{-1} \frac{d^2y}{dx^2} + (-1)\bar{y}^{-1-1} \cdot \frac{dy}{dx} \cdot \frac{dy}{dx} = \frac{d^2y}{dx^2}$$

$$\frac{1}{y} \cdot \frac{d^2y}{dx^2} - \frac{1}{y^2} \left(\frac{dy}{dx} \right)^2 = \frac{d^2y}{dx^2}$$

$$-\frac{1}{y^2} \left(\frac{dy}{dx} \right)^2 = \frac{d^2y}{dx^2} - \frac{1}{y} \frac{d^2y}{dx^2}$$

$$= \frac{d^2y}{dx^2} \left(1 - \frac{1}{y} \right)$$

$$\left(1 - \frac{1}{y} \right) \frac{d^2y}{dx^2} + \frac{1}{y^2} \left(\frac{dy}{dx} \right)^2 = 0$$

Ans

CHAPTER No. 9
EXERCISE NO 9.1

Q2 Find D.E of which
The given function is soln.

(iv) $y = a e^x + b \ln x + c x + d$. — (1)

Diff w.r.t x

$$\frac{dy}{dx} = a e^x + b \cdot \frac{1}{x} + c + 0$$

Now Diff again w.r.t x

$$\frac{d^2 y}{dx^2} = a e^x - \frac{b}{x^2} \quad \text{--- (2)}$$

Diff again w.r.t x

$$\frac{d^3 y}{dx^3} = a e^x + \frac{2b}{x^3} \quad \text{--- (3)}$$

Diff again w.r.t x

$$\frac{d^4 y}{dx^4} = a e^x - 2 \left(\frac{3}{x^4} \right) b$$

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$\frac{1}{x^3} = x^{-3}$$

$$-3x^{-4}$$

$$= -\frac{3}{x^4}$$

$$\frac{d^4 y}{dx^4} = a e^x - \frac{6b}{x^4} \quad \text{--- (4)}$$

Eliminate a, b from (2), (3), (4)

$$\begin{vmatrix} \frac{d^2 y}{dx^2} & e^x & -\frac{1}{x^2} \\ \frac{d^3 y}{dx^3} & e^x & \frac{2}{x^3} \\ \frac{d^4 y}{dx^4} & e^x & -\frac{6}{x^4} \end{vmatrix} = 0$$

taking e^x Common from C_1
and $\frac{1}{x^2}$ from C_3

$$e^x \cdot \frac{1}{x^2} \begin{vmatrix} \frac{d^2 y}{dx^2} & 1 & -1 \\ \frac{d^3 y}{dx^3} & 1 & \frac{2}{x} \\ \frac{d^4 y}{dx^4} & 1 & -\frac{6}{x^2} \end{vmatrix}$$

$$\frac{d^4 y}{dx^4} = a e^x - \frac{6b}{x^4} \quad \text{--- (4)}$$

Eliminate a, b from (2), (3), (4)

$$\begin{vmatrix} \frac{d^2 y}{dx^2} & e^x & -\frac{1}{x^2} \\ \frac{d^3 y}{dx^3} & e^x & \frac{2}{x^3} \\ \frac{d^4 y}{dx^4} & e^x & -\frac{6}{x^4} \end{vmatrix} = 0$$

taking e^x common from C_1
and $\frac{1}{x^2}$ from C_3

$$\left(e^x \cdot \frac{1}{x^2} \right) \begin{vmatrix} \frac{d^2 y}{dx^2} & 1 & -1 \\ \frac{d^3 y}{dx^3} & 1 & \frac{2}{x} \\ \frac{d^4 y}{dx^4} & 1 & -\frac{6}{x^2} \end{vmatrix} = 0$$

$$\begin{vmatrix} \frac{d^2 y}{dx^2} & 1 & -1 \\ \frac{d^3 y}{dx^3} & 1 & \frac{2}{x} \\ \frac{d^4 y}{dx^4} & 1 & -\frac{6}{x^2} \end{vmatrix} = 0$$

Expand from R_1

$$x(2+x) \frac{d^4 y}{dx^4} = (2x+6) \frac{d^3 y}{dx^3} - (6-x^2) \frac{d^2 y}{dx^2}$$

Ans.

CHAPTER No 9
EXERCISE No 9.2

Q2 (v) Find D.E of
 $x^2 + y^2 + 2gx + 2fy + C = 0$

Sol $x^2 + y^2 + 2gx + 2fy + C = 0$

Diff w.r.t x

$$2x + 2y \frac{dy}{dx} + 2g(1) + 2f \frac{dy}{dx} + 0 = 0$$

Divide by 2

$$x + \left(y \frac{dy}{dx}\right) + (2g) + f \frac{dy}{dx} = 0 \quad \text{--- (1)}$$

Diff w.r.t x

$$1 + y \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{dy}{dx} + 0 + f \frac{d^2y}{dx^2} = 0$$

$$1 + \left(y \frac{d^2y}{dx^2}\right) + \left(\frac{dy}{dx}\right)^2 + f \frac{d^2y}{dx^2} = 0 \quad \text{--- (2)}$$

Diff w.r.t x

$$0 + y \cdot \frac{d^3 y}{dx^3} + \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} + 2 \left(\frac{dy}{dx} \right) \left(\frac{d^2 y}{dx^2} \right) + f \frac{d^3 y}{dx^3} = 0$$

$$(y+f) \frac{d^3 y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} = 0 \quad \text{--- (3)}$$

$$\text{From (2)} \quad (y+f) \frac{d^2 y}{dx^2} + 1 + \left(\frac{dy}{dx} \right)^2 = 0$$

$$(y+f) \frac{d^2 y}{dx^2} = -1 - \left(\frac{dy}{dx} \right)^2$$

$$\left(y+f \right) = \frac{-1 - \left(\frac{dy}{dx} \right)^2}{\frac{d^2 y}{dx^2}}$$

Put in (3)

$$\left[\frac{-1 - \left(\frac{dy}{dx} \right)^2}{\frac{d^2 y}{dx^2}} \right] \frac{d^3 y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} = 0$$

$$\left[-1 - \left(\frac{dy}{dx} \right)^2 \right] \frac{d^3 y}{dx^3} + 3 \left(\frac{dy}{dx} \right) \left(\frac{d^2 y}{dx^2} \right)^2 = 0$$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right] \frac{d^3 y}{dx^3} - 3 \left(\frac{dy}{dx} \right) \left(\frac{d^2 y}{dx^2} \right)^2 = 0$$

CHAPTER No 9

EXERCISE No 9.2

Q2 (vi) $u = f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$

Sol

$$u = (x^2 + y^2 + z^2)^{-1/2} \quad \text{--- (1)}$$

Diff w.r.t. x Keeping

y, z Constant.

$$\frac{\partial u}{\partial x} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2 - 1} \cdot (2x + 0 + 0)$$

$$= -\frac{2x}{2} (x^2 + y^2 + z^2)^{-3/2}$$

$$\frac{\partial u}{\partial x} = -x (x^2 + y^2 + z^2)^{-3/2}$$

Diff again w.r.t. x

Keeping y, z Constant

$$\frac{\partial^2 u}{\partial x^2} = -1 \cdot (x^2 + y^2 + z^2)^{-3/2} - x \left(-\frac{3}{2}\right) (x^2 + y^2 + z^2)^{-3/2 - 1} \cdot (2x + 0 + 0)$$

$$\frac{\partial^2 u}{\partial x^2} = -(x^2 + y^2 + z^2)^{-3/2} + 3x^2 (x^2 + y^2 + z^2)^{-5/2} \quad \text{--- (2)}$$

Similarly Diff (1) w.r.t y two times keeping x, z Constant. we get -

$$\frac{\partial^2 u}{\partial y^2} = -(\underline{x^2 + y^2 + z^2})^{-\frac{3}{2}} + 3y^2(x^2 + y^2 + z^2)^{-\frac{5}{2}} \quad (3)$$

Similarly when we diff (1) two times w.r.t z, keeping x, y Constant we get

$$\frac{\partial^2 u}{\partial z^2} = -(\underline{x^2 + y^2 + z^2})^{-\frac{3}{2}} + 3z^2(x^2 + y^2 + z^2)^{-\frac{5}{2}} \quad (4)$$

adding (2), (3), (4)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -3(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3x^2(x^2 + y^2 + z^2)^{-\frac{5}{2}} + 3y^2(x^2 + y^2 + z^2)^{-\frac{5}{2}} + 3z^2(x^2 + y^2 + z^2)^{-\frac{5}{2}}$$

$$\begin{aligned} &= -3(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3(x^2 + y^2 + z^2)^{-\frac{5}{2}}(x^2 + y^2 + z^2) \\ &= -3(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3(x^2 + y^2 + z^2)^{-\frac{5}{2}+1} \\ &= -3(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3(x^2 + y^2 + z^2)^{-\frac{3}{2}} \end{aligned}$$

$$\left(\frac{3}{2}-1\right) \cdot (2x+0+0)$$

(2)

1

CHAPTER No 9.
EXERCISE No 9.2

Q2 (vii) $U = f(x-ay) + g(x+ay)$
where f and g are Differentiable Twice.

Sol $U = f(x-ay) + g(x+ay) \text{ --- (1)}$

Diff w.r.t. x keeping y Constant

$$\frac{\partial U}{\partial x} = f'(x-ay)(1-0) + g'(x+ay)(1+0)$$

$$\frac{\partial U}{\partial x} = f'(x-ay) + g'(x+ay)$$

Diff again w.r.t. x

$$\frac{\partial^2 U}{\partial x^2} = f''(x-ay) \cdot 1 + g''(\underline{x+ay}) \cdot 1$$

$$\frac{\partial^2 U}{\partial x^2} = f''(x-ay) + g''(x+ay) \text{ --- (2)}$$

Again take (1)

$$u = f(x - ay) + g(x + ay) \quad \text{--- (1)}$$

Diff w.r.t. y keeping x constant

$$\frac{\partial u}{\partial y} = f'(x - ay)(0 - a) + g'(x + ay)(0 + a)$$

$$= -af'(x - ay) + ag'(x + ay)$$

Diff again w.r.t. y
keeping x constant

$$\frac{\partial^2 u}{\partial y^2} = -[af''(x - ay)](0 - a) + [ag''(x + ay)](0 + a)$$

$$= a^2 f''(x - ay) + a^2 g''(x + ay)$$

$$= a^2 [f''(x - ay) + g''(x + ay)] \quad \text{--- (3)}$$

$$\frac{\partial^2 u}{\partial y^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad \text{using (2)}$$

Q3 Find The D.E of all circles of radius "a"

Sol Let $C(h, k)$ be the center of a circle of radius "a" Then Equation

$$(x-h)^2 + (y-k)^2 = a^2 \quad \text{--- (1)}$$

This Equation contain two arbitrary Constant h, k
Therefore we have to diff (1) two times

Diff (1) w.r.t. x

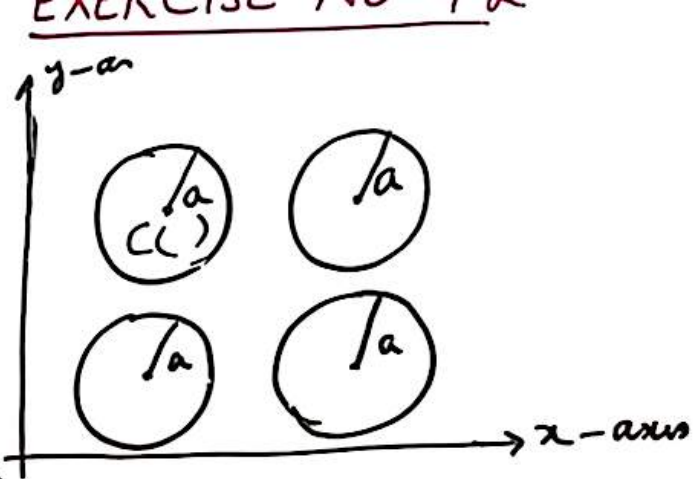
$$2(x-h) \cdot (1-0) + 2(y-k) \cdot (y'-0) = 0$$

$$2(x-h) + 2(y-k)y' = 0$$

Divide by 2

$$(x-h) + (y-k)y' = 0 \quad \text{--- (2)}$$

Diff (2) w.r.t. x



$$1 - 0 + (y - k)y'' + (y' - 0)y' = 0$$

$$1 + (y - k)y'' + y'^2 = 0$$

$$(y - k)y'' = -1 - y'^2$$

$$y - k = \frac{-1 - y'^2}{y''} = -\left(\frac{1 + y'^2}{y''}\right)$$

Put in (2)

$$(x - h) + \left(\frac{-1 - y'^2}{y''}\right)y' = 0$$

$$(x - h) - \left(\frac{1 + y'^2}{y''}\right)y' = 0$$

$$(x - h) = \left(\frac{1 + y'^2}{y''}\right)y' \quad (4)$$

Put (3), (4) in (1)

$$\left(\frac{1 + y'^2}{y''}\right)^2 \cdot y'^2 + \left(\frac{1 + y'^2}{y''}\right)^2 = a^2 \quad \text{multiply by } (y'')^2$$

$$(1 + y'^2)^2 \cdot y'^2 + (1 + y'^2)^2 = a^2 \cdot (y'')^2$$

$$\frac{dy}{dx} = y'$$

$$\frac{d^2y}{dx^2} = y''$$

Q3(ii) Find the D.E of
circle passes through origin.

Sol Equation of the circle passes through origin is

$$x^2 + y^2 + 2gx + 2fy = 0 \quad \text{--- (1)}$$

Diff (1) w.r.t. x

$$2x + 2yy' + 2g \cdot 1 + 2fy' = 0 \quad \text{--- (2)}$$

Diff (2) w.r.t. x

$$2 + 2(yy'' + y'y') + 0 + 2fy'' = 0$$

$$2 + 2(yy'' + y'^2) + 2fy'' = 0$$

Div by 2.

$$1 + (yy'' + y'^2) + fy'' = 0 \quad \text{--- (3)}$$

multiply (2) by x

$$2x^2 + 2xyy' + (2gx) + 2xy'f = 0 \quad \text{--- (4)}$$

$$x^2 + y^2 + (2gx) + 2fy = 0 \quad \text{--- (1)}$$

Subtracting (1) from (4)

$$x^2 + 2xyy' + \underline{2xy'f} - y^2 - \underline{2fy} = 0$$

$$(2xy'f - 2fy) = y^2 - x^2 - 2xyy'$$

$$f(2xy' - 2y) = y^2 - x^2 - 2xyy'$$

$$f = \frac{y^2 - x^2 - 2xyy'}{2xy' - 2y} \quad \text{Put in (3)}$$

$$1 + (yy'' + y'^2) + \left(\frac{y^2 - 2xyy'}{2xy' - 2y} \right) y'' = 0 \quad \text{Ans}$$

multiply by $(2xy' - 2y)$

$$(2xy' - 2y) + (2xy' - 2y)(yy'' + y'^2) + (y^2 - 2xyy')y'' = 0$$

Q3iii D.E of all Ellipse in standard form.

Sol Standard form of ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

Diff w.r.t x

$$\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$$

Div by 2

$$\frac{x}{a^2} + \frac{yy'}{b^2} = 0 \quad \text{--- (2)}$$

Diff w.r.t x

$$\frac{1}{a^2} + \frac{1}{b^2} [yy'' + y' \cdot y'] = 0$$

$$\frac{1}{a^2} + \frac{1}{b^2} [yy'' + y'^2] = 0 \quad \text{--- (3)}$$

multiply (3) by x

$$\left(\frac{x}{a^2}\right) + \frac{x}{b^2} [yy'' + y'^2] = 0 \quad \text{--- (4)}$$

$$\left(\frac{x}{a^2}\right) + \frac{1}{b^2} yy' = 0 \quad \text{--- (2)}$$

subtracting (2) from (4)

$$\frac{x}{b^2} [yy'' + y'^2] - \frac{1}{b^2} yy' = 0$$

$$\left(\frac{1}{b^2}\right) [xyy'' + xy'^2 - yy'] = 0$$

$$\frac{1}{b^2} \neq 0$$

$$xyy'' + x\dot{y}^2 - yy' = 0$$

is the Required D.E.

Q3 (iv), D.E of Parabola which has latus rectum $(4a)$ and axis are // to x -axis

Sol The Required. Eq of Parabola is of the form.

$$(y-k)^2 = 4a(x-h) \text{ --- (1)}$$

Diff ① w.r.t. x

$$2(y-k)^{2-1} \cdot (y' - 0) = 4a(1-0)$$

$$2(y-k)(y') = 4a$$

Divide by 2

$$y'(y-k) = 2a \text{ --- (2)}$$

Diff w.r.t. x

$$y''(y-k) + y'(y'-0) = 0$$

$$y''(y-k) + y'^2 = 0$$

$$y''(y-k) = -y'^2$$

Divide by y''

$$y-k = -\frac{y'^2}{y''}$$

Put in (2)

$$y' \left(-\frac{y'^2}{y''} \right) = 2a$$

multiply by y''

$$-y'^3 = 2a y''$$

$$y'^3 + 2a y'' = 0$$

$$\left(\frac{dy}{dx} \right)^3 + 2a \frac{d^2y}{dx^2} = 0$$

Q3(V) Find D.E of Hyperbola
in standard form.

Sol Standard form of Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

This Eq contains two arbitrary Constant.
 $\therefore$ we have to diff (1) two times.

Diff (1) w.r.t. x

$$\frac{2x}{a^2} - \frac{2y^{2-1} \cdot y'}{b^2} = 0$$

Divide by 2

$$\frac{x}{a^2} - \frac{yy'}{b^2} = 0 \quad \text{--- (2)}$$

Diff again w.r.t. x

$$\frac{1}{a^2} - \frac{1}{b^2} [yy'' + y'y'] = 0 \quad \text{--- (3)}$$

multiply by x

$$\frac{x}{a^2} - \frac{1}{b^2} [xyy'' + xy'^2] = 0 \quad \text{--- (4)}$$

$$\frac{x^2}{a^2} - \frac{1}{b^2} [yy'] = 0 \quad \text{--- (2)}$$

Subtract (2) from (4)

$$-\frac{1}{b^2} [xyy'' + xy'^2] + \frac{1}{b^2} (yy') = 0$$

$$-\frac{1}{b^2} [(xyy'' + xy'^2) - yy'] = 0$$

$$-\frac{1}{b^2} \neq 0$$

$$xyy'' + xy'^2 - yy' = 0$$

$$xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$$

EXERCISE NO. 1 & 2

Q3(V) Find D.E of Conics whose axis coincides with axis of coordinate.

Sol Equation of Conics is

$$ax^2 + by^2 = 1 \text{ --- (1)}$$

This Equation contains two arbitrary constants a and b . Therefore we have to diff (1) two times

Diff (1) w.r.t. x

$$2ax + b(2y^{2-1} \cdot \frac{dy}{dx}) = 0$$

$$(2)ax + (2)by \frac{dy}{dx} = 0$$

Dividing by 2 $\neq 0$

$$ax + by \frac{dy}{dx} = 0 \text{ --- (2)}$$

Subb ② w.r.t. x

$$a(1) + b \left[y \cdot \frac{d^2 y}{dx^2} + \frac{dy}{dx} \cdot \frac{dy}{dx} \right] = 0$$

multiply x

$$\textcircled{ax} + b \left[xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 \right] = 0 \quad \textcircled{3}$$

$$\textcircled{ax} + by \frac{dy}{dx} = 0 \quad \text{---} \quad \textcircled{2}$$

Subtract ② from ③

$$b \left[xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 \right] - by \frac{dy}{dx} = 0$$

$$b \left[xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right) \right] = 0$$

$$b \neq 0$$

$$xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right) = 0$$